SIMATS ENGINEERING
ASSESSMENT TOOL 1

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Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

   Given:
   The outcomes are:
   H H T H H H T H H T
   Total number of tosses (**n**) = **10**
   Number of Heads (**H**) = **7**
   Number of Tails (**T**) = **3**
   We need to estimate the probability of getting Heads (**p**) using **Maximum Likelihood Estimation (MLE).**

   Step 1: Formula
   For a Bernoulli/Binomial experiment,
   $$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Number of Trials}}$$

   Here,
   Success = Getting Heads
   Therefore,
   $$\hat{p} = \frac{7}{10}$$

   Step 2: Calculate
   $$\hat{p} = 0.7$$

   Final Answer
   Estimated probability of getting Heads (MLE):
   $$\boxed{\hat{p} = 0.7}$$
   Therefore MLE chooses the probability value that makes the observed data most likely. Since Heads occurred **7 out of 10 times**, the best estimate is **0.7**.

2. Out of 50 students, 42 passed an exam.
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
   Given
   Total students (**n**) = **50**
   Students passed = **42**
   Students failed = **8**
   Need to estimate the probability of passing.

   Step 1: Formula
   For Bernoulli distribution,

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Trials}}$$

   Here,
   Success = Passing
   Therefore,

$$\hat{p} = \frac{42}{50}$$

   Step 2: Calculate

$$\hat{p} = 0.84$$

   Final Answer

$$\boxed{\hat{p} = 0.84}$$

   Therefore,According to MLE, the best estimate of passing probability is the observed proportion. Since **42 students passed**, the estimated probability is **84%**.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.
   Given
   Sample size = **100 bulbs**
   Defective bulbs = **8**
   Non-defective bulbs = **92**
   Need to estimate the probability of a bulb being defective.
   Step 1: Formula

$$\hat{p} = \frac{\text{Number of Defective Bulbs}}{\text{Total Bulbs}}$$

   Step 2: Substitute

$$\hat{p} = \frac{8}{100}$$

   Step 3: Calculate

$$\hat{p} = 0.08$$

Final Answer

$$\boxed{\hat{p} = 0.08}$$

Therefore,The observed defective rate is **8 out of 100**, so the MLE estimate is **0.08**.

4. A biased die is rolled 60 times. The number 6 appears 18 times.
   Estimate the probability of rolling a 6 using MLE.

   Given
   Total rolls = **60**
   Number of times 6 appeared = **18**
   Need to estimate probability of rolling a 6.
   Step 1: Formula

   $$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Trials}}$$

   Success = Rolling 6

   Step 2: Substitute

   $$\hat{p} = \frac{18}{60}$$

   Step 3: Calculate

   $$\hat{p} = 0.3$$

   Final Answer

   $$\boxed{\hat{p} = 0.3}$$

   Therefore, Since the number **6** appeared **18 times** out of **60**, the estimated probability is **0.3**.

5. Given:
   Input (x) = 4
   Weight (w) = 2
   Actual Output = 10
   Prediction:
   y = w × x
   1. Find the predicted output.
   2. Calculate the error.
   3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Given
Input,

$$x = 4$$

Weight,

$$w = 2$$

Actual Output,

$$y_{actual} = 10$$

Prediction Formula,

$$y = w \times x$$

Gradient,

$$\frac{\partial L}{\partial w} = 4$$

Learning Rate,

$$\alpha = 0.01$$

Need to find

1. Predicted Output
2. Error
3. Updated Weight

Step 1: Predicted Output
Formula

$$y = w \times x$$

Substitute values

$$y = 2 \times 4$$
$$y = 8$$

Predicted Output

$$\boxed{8}$$

Step 2: Error
Formula

$$\text{Error} = Actual - Predicted$$

Substitute values

$$= 10 - 8$$
$$= 2$$

Error

$$\boxed{2}$$

Step 3: Update the Weight
Gradient Descent Formula

$$w_{new} = w_{old} - \alpha \times Gradient$$

Substitute values

$$w_{new} = 2 - (0.01 \times 4)$$

First calculate

$$0.01 \times 4 = 0.04$$

Updated weight,

$$w_{new} = 2 - 0.04$$
$$w_{new} = 1.96$$